

Project Summary

An outbreak of Foot and Mouth Disease and many other emergency animal diseases (EAD) in Australia is likely to infect feral animals of domestic species origin. In Australia, animals susceptible to FMD infection and other EAD's include both intensively and extensively managed ungulate species as well as their wild/feral/unmanageable counterparts. Such species include cattle, sheep, pigs, goats, buffalo and camel.

In its June 2002 research report, 'Impact of a foot and mouth disease outbreak in Australia', the Productivity Commission examined in detail the potential social, economic and environmental consequences of an FMD outbreak. The Commission's worst-case scenario involved key beef and lamb export markets being closed for 15 months. The cost of a foot and mouth disease incursion under this scenario would be between \$8 billion and \$13 billion of gross domestic product and its consequences would be felt for nearly 10 years after the event. Even an isolated outbreak that was brought rapidly under control was estimated to potentially cost \$2 to \$3 billion of gross domestic product.

An eradication or control program for an EAD incursion would be conducted under AUSVETPLAN and EADRA (Emergency Animal Disease Response Agreement) guidelines and may require the culling of feral and unmanageable animals in designated areas as part the response program.

In many instances where culling was required, this would involve the destruction of these animals in situ. That is, these animals are not able to be managed in the conventional sense, for example, mustered and yarded, and so need to be destroyed by methods that do not require them to be managed before destruction. Such methods include aerial shooting, poisoning, trapping in small numbers for destruction. These destruction methods are well recognized and are used routinely for other purposes during the normal management of the environment these animals frequent and at this point do not need further development.

However, destruction using these methods will leave carcasses scattered in situ through an area, usually wherever an animal falls following destruction. AUSVETPLAN Disposal Manual gives a range of disposal options for carcasses that are suitable for animals that can be controlled and managed before destruction. For example, mustered, yarded and then destroyed in a site that allows ready access for machinery to undertake carcass disposal. The AUSVETPLAN Disposal Manual does not give options suitable for wild/feral/unmanageable animals that fall in situ after destruction over a wide area, especially considering that the exact location of each carcass may not be known, the site may be impractical and inaccessible to reach and the application of existing options very expensive and impractical.

As well as assessing the effectiveness of normal decomposition processes to control EAD viruses, natural predation on carcasses also needs to be considered. Predation forms a possible route for EAD infection to transfer from carcasses destroyed and left in situ to scavenging animals, most notably pigs but other species eg cattle are also known to scavenge carcasses. There is then an ensuing risk that an ongoing cycle of infection may occur in the feral population causing an ongoing reservoir of the EAD. It is known that predation occurs on isolated carcasses in rural and extensive areas, however it is not

known/confirmed how much predation will occur in an area where large numbers of a feral population have been destroyed.

As part of its EAD preparedness planning, Biosecurity Queensland (BQ) is assessing how this situation can be managed during a response. A preliminary trial conducted by BQ in association with a helicopter feral pig cull conducted at a National Park in Western Queensland, on 25 th August 2006 gave promising results and showed pH changes consistent with FMD virus destruction (from internal report of Williamson, G., 2006). These post mortem changes are also likely to be effective against a range of other EAD viruses as well. The impact of predation was not examined during this trial.

This further trial is proposed to confirm the preliminary trial finding and to assess the rate of predation on carcasses in areas where large numbers of animals have been destroyed. The results of these trials would be used to allow assessment of the technique for inclusion into AUSVETPLAN to add to the options that already exist there.

Objectives

- Assess pH and temperature changes in decomposing carcasses
- Assess species and environmental effects on carcass degradation in relation to pH and temperature
- Assess the level of predation that occurs in an area where large numbers of animals have been destroyed
- Validate “destroy and let lie” method as an acceptable disposal method that will inactivate a range of EAD viruses
- If validated, supply information for assessment for the update of AUSVETPLAN.

Final Report

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